Optimal Control of the Inverted Pendulum

Solving the Algebraic Riccati Equation via Newton's Method

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1 System Model

The inverted pendulum is linearized around the unstable upright equilibrium. The dynamics are represented in state-space form:

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + B\boldsymbol{u} \tag{1}$$

with state vector

$$\boldsymbol{x} = \begin{bmatrix} \theta \\ \dot{\theta} \\ z \\ \dot{z} \end{bmatrix}. \tag{2}$$

Here, θ denotes the pendulum angle and z denotes the cart position.

2 LQR Problem Formulation

The objective is to minimize the infinite-horizon quadratic cost

$$J = \int_0^\infty \left(\boldsymbol{x}^T Q \boldsymbol{x} + \boldsymbol{u}^T R \boldsymbol{u} \right) dt, \tag{3}$$

where $Q \succeq 0$ and $R \succ 0$.
The optimal feedback law is

$$\boldsymbol{u}^* = -K\boldsymbol{x}, \qquad K = R^{-1}B^T P, \tag{4}$$

where P is the positive-definite solution of the Algebraic Riccati Equation.

3 Algebraic Riccati Equation

Definition 1 (Algebraic Riccati Equation)**.** *The continuous-time Algebraic Riccati Equation (ARE) is*

$$A^T P + PA - PBR^{-1}B^T P + Q = 0. \tag{5}$$

Equation (5) is nonlinear because the unknown matrix P appears quadratically.

Under standard stabilizability and detectability assumptions, there exists a unique positive-definite stabilizing solution P^*.

4 Newton's Method for the ARE

Newton's method solves the nonlinear matrix equation iteratively.

4.1 Residual Function

Define

$$\mathcal{R}(P) = A^T P + PA - PBR^{-1}B^T P + Q. \tag{6}$$

The goal is to find P^* such that

$$\mathcal{R}(P^*) = 0.$$

4.2 Newton Iteration

Given the current iterate P_k, define

$$A_k = A - BR^{-1}B^T P_k. \tag{7}$$

The Newton correction ΔP_k is obtained by solving

$$A_k^T \Delta P_k + \Delta P_k A_k = -\mathcal{R}(P_k). \tag{8}$$

Equation (8) is a Lyapunov equation.

The update is

$$P_{k+1} = P_k + \Delta P_k. \tag{9}$$

4.3 Stopping Criterion

Iterations are terminated when

$$\|P_{k+1} - P_k\|_F < 10^{-8}. \tag{10}$$

Theorem 2 (Quadratic Convergence)**.** *If the initial guess is sufficiently close to the stabilizing solution P^*, Newton's method converges quadratically:*

$$\|P_{k+1} - P^*\|_F \leq c\,\|P_k - P^*\|_F^2,$$

for some constant $c > 0$.

5 Numerical Considerations

Remark 3 (Symmetry Enforcement). *To avoid loss of symmetry due to numerical round-off,*

$$P_k \leftarrow \frac{P_k + P_k^T}{2}.$$

Remark 4 (Initial Guess). *A common choice is*

$$P_0 = Q.$$

A stabilizing initial guess generally improves convergence.

Remark 5 (Tolerance). *A tolerance of* 10^{-8} *typically provides an excellent compromise between numerical accuracy and computational effort.*

6 Newton's Method — The General Idea

For a scalar equation f(x)=0, Newton's update is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

For a matrix equation F(P) = 0 , the same idea applies but the "derivative" becomes a Fréchet derivative — a linear operator acting on matrix perturbations.

$$\text{Step 1 — Define the Residual:} \quad \mathcal{F}(P) = A^T P + PA - PBR^{-1}B^T P + Q$$

$$\text{We want:} \quad \mathcal{F}(P^*) = 0$$

Step 2 — Compute the Fréchet Derivative We perturb P by a small matrix P:

$$\mathcal{F}(P + \Delta P) = A^T(P + \Delta P) + (P + \Delta P)A - (P + \Delta P)BR^{-1}B^T(P + \Delta P) + Q$$

First-order expansion (neglecting PP):

$$\mathcal{F}(P + \Delta P) \approx \mathcal{F}(P) + A^T \Delta P + \Delta PA - \Delta PBR^{-1}B^T P - PBR^{-1}B^T \Delta P$$

where:

$$\mathcal{F}(P) = A^T P + PA - PBR^{-1}B^T P + Q$$

Fréchet derivative form:

$$D\mathcal{F}(P)[\Delta P] = A^T \Delta P + \Delta PA - \Delta PBR^{-1}B^T P - PBR^{-1}B^T \Delta P$$

Step 3— Write the Newton Equation The Newton step solves:

$$D\mathcal{F}(P_k)[\Delta P] = -\mathcal{F}(P_k)$$

Expanding the Fréchet derivative :

$$A^T \Delta P + \Delta PA - \Delta PBR^{-1}B^T P_k - P_k BR^{-1}B^T \Delta P = -\mathcal{F}(P_k)$$

Define the closed-loop matrix:

$$A_k = A - BR^{-1}B^T P_k$$

by Substitution, we obtain the Lyapunov form

$$A_k^T \Delta P + \Delta P A_k = -\mathcal{F}(P_k)$$

Then the result(final boxed result:

$$\boxed{A_k^T \Delta P + \Delta P A_k = -\mathcal{F}(P_k)}$$

Step 4 — Newton–Raphson Iteration Scheme Once P is computed from the Lyapunov equation, we up

$$P_{k+1} = P_k + \Delta P$$

Why Lyapunov and Not Direct Inversion?

Because the Newton equation

$$D\mathcal{F}(P)[\Delta P] = -\mathcal{F}(P)$$

is not a scalar equation of the form

$$\frac{d}{dP}\mathcal{F}(P)\,\Delta P = -\mathcal{F}(P).$$

The derivative $D\mathcal{F}(P)$ is a **Fréchet derivative**, i.e., a linear operator acting on matrices, not a scalar derivative.

The unknown ΔP is an $n \times n$ matrix, so the problem cannot be solved using direct inversion in the usual scalar sense.

References

1. F. L. Lewis, D. Vrabie, and V. L. Syrmos, *Optimal Control*, 3rd Edition, Wiley, 2012.

2. B. D. O. Anderson and J. B. Moore, *Optimal Control: Linear Quadratic Methods*, Dover, 2007.

   (a) https://github.com/Drlecteur19/inverted-pendulum-lqr-control